Last time:

Stochastic integral: Brownian Motion

- Definition of $\int_0^t \varphi_u \, dW_u$

  as a limit (in $L^2(\mathbb{P})$) of sums of form $\sum \varphi_{t_i}(W_{t_{i+1}} - W_{t_i})$

- Properties of Stoch. Int.:

$$I = \int_0^t \varphi_u \, dW_u$$

  I linear, continuous, adapted, Itô isometry

$\rightsquigarrow$ Stratonovich Integral.

---

§4. Stochastic Calculus

So far, we've defined

$$X_t = X_0 + \int_0^t \alpha_u \, du + \int_0^t \varphi_u \, dW_u$$

for $(\alpha_u), (\varphi_u)$ s.t. $\int |\alpha_u| \, du < \infty$

and $\mathbb{E} \int_0^t \varphi_u^2 \, du < \infty$.

Weaken this slightly:

if $\mathbb{P}\left(\int_0^t \varphi_u^2 \, du < \infty, \ \forall t \geq 0\right) = 1$ (12)

then consider $\varphi_t^N := \begin{cases} \varphi_t & \int_0^t \varphi_u^2 \, du \leq N \\ 0 & \text{o/wise} \end{cases}$

Then $\lim_{N \to \infty} \int_0^t \varphi_u^N \, dW_u$ is well defined

if $\int_0^t \varphi_u^2 \, du \leq N$, some N

since $\int_0^t \varphi_u^N \, dW_u = \int_0^t \varphi_u^{N'} \, dW_u$

if $\int_0^t \varphi_u^2 \, du \leq \max\{N, N'\}$

By (12) this is a set with probability 1, so can define $\int_0^t \varphi_u \, dW_u$ if (12) holds.

Let the set of adapted processes satisfying (12) be $\mathcal{H}$,

$\hat{\mathcal{H}}$ set of processes s.t.

$$\mathbb{P}\left(\int_0^t |\alpha_u| \, du < \infty \ \forall t \geq 0\right) = 1$$

<u>Def^n</u>: A (one-dimensional) Itô process is a stochastic process on $\mathbb{R}$ of the form:

$$X_t = X_0 + \int_0^t \alpha_u \, du + \int_0^t \varphi_u \, dW_u \quad (14)$$

$$\alpha \in \hat{\mathcal{H}}, \ \varphi \in \mathcal{H}.$$

Eg. $W_t \in \mathcal{H}$, $\frac{1}{2} \in \hat{\mathcal{H}}$ so

$$\frac{W_t^2}{2} = \frac{W_0^2}{2} + \int_0^t \tfrac{1}{2} \, du + \int_0^t W_u^2 \, dW_u$$

$\Rightarrow \frac{W_t^2}{2}$ an Itô process

<u>Def^n</u>: Let X_t be an Itô process as in (14). Then if ψ_u in

$$\mathcal{H}[X] := \left\{ \psi : \int_0^t |\alpha_u \psi_u| \, du + \int_0^t (\varphi_u \psi_u)^2 \, du < \infty \ \ \mathbb{P}\text{-a.s. } \forall t \geq 0 \right\}$$

Then $Y_t := \int_0^t \psi_u \, dX_u$

$$\int_0^t dY_u + \underset{\overset{\shortparallel}{0}}{Y_0} = \int_0^t (\alpha_u \psi_u) \, du + \int_0^t (\varphi_u \psi_u) \, dW_u$$

<u>OR</u>:

$$\begin{aligned} dY_t &= \psi_t \, dX_t \\ &= \psi_t (\alpha_t \, dt + \varphi_t \, dW_t) \\ &= \psi_t \alpha_t \, dt + \psi_t \varphi_t \, dW_t \end{aligned}$$

<u>Note</u>: Also can show:

$$Y_t = \lim \sum \psi_{t_i} (X_{t_{i+1}} - X_{t_i})$$

<u>Corollary 4.4</u>: Let X_t be an Itô process, $\psi \in \mathcal{H}[X]$, $Y = \int \psi_u \, dX_u$

Then Y is an Itô process, and if $\xi \in \mathcal{H}[Y]$, then $\xi_u \psi_u \in \mathcal{H}[X]$

and $\int \xi_u \, dY_u = \int \xi_u \psi_u \, dX$

§4.1 Itô's Lemma:

Theorem 4.5 (Itô's Lemma)

Let X_t be an Itô process, given by (14), suppose $g \in C^2([0,\infty) \times \mathbb{R})$.

Then $Y_t = g(t, X_t)$ is an Itô process

$$dY_t = \left(\frac{\partial g}{\partial t}(t, X_t) + \alpha_t \frac{\partial g}{\partial x}(t, X_t) + \frac{1}{2} \varphi_t^2 \frac{\partial^2 g}{\partial x^2}(t, X_t) \right) dt + \varphi_t \frac{\partial g}{\partial x}(t, X_t)\, dW_t \qquad (15)$$

Note from Thm 2.6,

"roughly": $(dW_t) \sim (dt)^{1/2}$

$$(dW_t)^2 \sim dt$$

so $dW_t\, dt \sim (dt)^{3/2} \sim (dW_t)^3$

so
$$\begin{aligned}(dX_t)^2 &= (\alpha_t\, dt + \varphi_t\, dW_t)^2 \\ &= \alpha_t^2 \cancel{(dt)^2}^{\to 0} + 2\alpha_t \varphi_t \cancel{dt\, dW_t}^{\to 0} + \varphi_t^2 (dW_t)^2 \\ &= \varphi_t^2\, dt\end{aligned}$$

And (15) is what we get when we expand

$$dY_t = \frac{\partial g}{\partial t} dt + \frac{\partial g}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 g}{\partial x^2} (dX_t)^2$$

which is a Taylor Expansion of $g(t, X_t)$

Sketch Proof

By various approx. arguments, assume g is C^3, bdd derivatives up to order 3, φ_t, α_t are bdd and simple.

Let Π^n be a sequence of partitions $0 = t_0^n \le t_1^n \le \dots \le t_{m_n}^n = t$ s.t. $\|\Pi^n\| \to 0$

write $\alpha_{j,n} := \alpha_{t_j^n}$, $\Delta t_{j,n} = t_{j+1}^n - t_j^n$ etc.

Then Taylor's Thm implies:

$$g(t, X_t) = g(0, X_0) + \sum_{j=0}^{m_n - 1} \left(g(t_{j+1,n}, X_{j+1,n}) - g(t_{j,n}, X_{j,n}) \right)$$

$$= g(0, X_0) + \sum_{j=0}^{m_n-1} \frac{\partial g}{\partial t}(t_{j,n}, X_{j,n}) \Delta t_{j,n}$$

$$+ \sum_j \frac{\partial g}{\partial x}(\cdot) \Delta X_{j,n}$$

$$+ \frac{1}{2} \sum \left[\frac{\partial^2 g}{\partial t^2} (\Delta t_{j,n})^2 + 2 \frac{\partial^2 g}{\partial x \partial t} (\Delta t_{j,n} \Delta X_{j,n}) + \frac{\partial^2 g}{\partial x^2} (\Delta X_{j,n})^2 \right]$$

$$+ \sum_j R_j$$

where $R_j = o(|\Delta t_{j,n}|^2 + |\Delta X_{j,n}|^2)$

Letting $n \to \infty$, first terms converge (in $\mathbb{P}$) $\int \frac{\partial g}{\partial t} dt$ and $\int_0^t \frac{\partial g}{\partial x} dX_t$.

Also $\sum R_j \longrightarrow 0$

Since $\frac{\partial^2 g}{\partial t^2}$ is bounded, so

$$\sum \left| \frac{\partial^2 g}{\partial t^2} (\Delta t_{j,n})^2 \right| \le g^* \|\Pi^n\| t \longrightarrow 0$$

and

$$\eta(s) := \sum \frac{\partial^2 g}{\partial x \partial t} (\Delta t_{j,n}) \mathbb{1}_{[t_{j,n}, t_{j+1,n})}(s) \longrightarrow 0$$

so

$$\sum \eta(t_{j,n}) \Delta X_{t_{j,n}} = \int \eta(s) \, dX_s = 0 \longrightarrow 0$$

Remaining term $\sum \frac{\partial^2 g}{\partial x^2} (\Delta X_{j,n})^2$

But assumed φ, α simple.

$$\Delta X_{j,n} = (\alpha_{j,n} \Delta t_{j,n} + \varphi_{j,n} \Delta W_{j,n})^2$$

$$\Rightarrow \sum \frac{\partial^2 g}{\partial x^2} (\Delta X_{j,n})^2 =$$

$$\sum \frac{\partial^2 g}{\partial x^2} \alpha_{j,n}^2 (\Delta t_{j,n})^2$$

$$+ 2 \sum \frac{\partial^2 g}{\partial x^2} \alpha_{j,n} \varphi_{j,n} (\Delta t_{j,n}) (\Delta W_{j,n})$$

$$+ \sum \frac{\partial^2 g}{\partial x^2} \varphi_{j,n}^2 (\Delta W_{j,n})^2$$

But $\frac{\partial^2 g}{\partial x^2}$, α, φ bdd $\Rightarrow$ first term, 2^{nd} term $\to 0$ (as above)

Write $\gamma_t = \frac{\partial^2 g}{\partial x^2}(t, X_t) \varphi_t^2$

NTS: $\sum \gamma_{j,n} (\Delta W_{j,n})^2 \longrightarrow \int_0^t \gamma_u \, du$

Recalling Thm 2.6:

$$\mathbb{E}\left[\left(\sum \gamma_{j,n} (\Delta W_{j,n})^2 - \sum \gamma_{j,n} \Delta t_{j,n} \right)^2 \right]$$

$$= \sum_{i,j} \mathbb{E}\left[\gamma_{i,n} \gamma_{j,n} (\Delta W_{i,n}^2 - \Delta t_{i,n}) \times (\Delta W_{j,n}^2 - \Delta t_{j,n}) \right]$$

(c.f. Thm 2.6)

$$= \sum_i \mathbb{E}\left[\gamma_{i,n}^2 (\Delta W_{i,n}^2 - \Delta t_{i,n})^2 \right]$$

$$= \sum \mathbb{E}[\gamma_{i,n}^2] \, \underbrace{\mathbb{E}\left[(\Delta W_{i,n}^2 - \Delta t_{i,n})^2 \right]}_{0}$$

Which is what we want

since

$$\sum_j \gamma_{j,n} t_{j,n} \longrightarrow \int_0^t \gamma_s \, ds$$

□

E.g. $Y_t = W_t^2$, $f(t,x) = x^2$

$$dY_t = 2W_t \, dW_t + \frac{1}{2} \cdot 2 \cdot (dW_t)^2$$

$$= 2W_t \, dW_t + dt$$

$$\int \Rightarrow Y_t - Y_0 = W_t^2 - W_0^2$$

$$= 2\int_0^t W_s \, dW_s + \underbrace{\int_0^t 1 \, du}_{= t}$$

$$\frac{W_t^2}{2} = \frac{W_0^2}{2} + \int_0^t W_s \, dW_s + \frac{1}{2} t.$$

---

§4.2 Integration by Parts:

X_t, Y_t Itô processes.

$$dX_t = \alpha_t \, dt + \varphi_t \, dW_t$$

$$dY_t = \beta_t \, dt + \psi_t \, dW_t$$

Then $d(X_t Y_t) = X_t \, dY_t + Y_t \, dX_t + dX_t \, dY_t$

Where $dX_t \, dY_t$ computed using

$$(dW_t)^2 = dt, \quad dW_t \, dt = dt^2 = 0$$

In particular:

$$X_t Y_t = X_0 Y_0 + \int_0^t X_s \, dY_s + \int_0^t Y_s \, dX_s + \underbrace{\int_0^t \varphi_s \psi_s \, ds}$$

§5. Stochastic Differential Equations (SDEs)

X_t solves the SDE

(21) $$dX_t = \mu(t, X_t)\,dt + \sigma(t, X_t)\,dW_t$$
$$X_0 = x_0$$ may be random

if $\quad X_t = x_0 + \int_0^t \mu(s, X_s)\,ds + \int_0^t \sigma(s, X_s)\,dW_s$

§5.1 Examples

Ex 5.1 (Geometric B.M.)

Fix $\mu \in \mathbb{R}$, $\sigma > 0$, $x_0 > 0$

Define $X_t = x_0 \exp\{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\}$

So $X_t = f(t, W_t)$

where $f(t,x) = x_0 \exp\{(\mu - \frac{1}{2}\sigma^2)t + \sigma x\}$

$\frac{\partial f}{\partial x} = \sigma f$, $\frac{\partial^2 f}{\partial x^2} = \sigma^2 f$, $\frac{\partial f}{\partial t} = (\mu - \frac{1}{2}\sigma^2) f$

So by Itô's Lemma:

$$dX_t = df(t, W_t) = (\mu - \tfrac{1}{2}\sigma^2) f(t, W_t)\,dt + \sigma f(t, W_t)\,dW_t + \tfrac{1}{2}\sigma^2 f(t, W_t)\underbrace{(dW_t)^2}_{=dt}$$

$$= \mu X_t\,dt + \sigma X_t\,dW_t$$

So X_t solves SDE with $\sigma(x) = \sigma x$
$\mu(x) = \mu x$.

Ex 5.2 (Ornstein-Uhlenbeck)

Let $\lambda, \sigma \in \mathbb{R}$. Consider

$$Y_t = \sigma \int_0^t e^{\lambda s}\,dW_s$$

Then $dY_t = \sigma e^{\lambda t} dW_t$

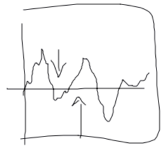
Now consider $X_t = e^{-\lambda t} Y_t$

By Itô's Lemma $\left(f(t,y) = e^{-\lambda t} y \right)$

$$dX_t = e^{-\lambda t} dY_t - \lambda e^{-\lambda t} Y_t \, dt$$
$$= -\lambda X_t \, dt + \sigma \, dW_t$$

So $X_t = e^{-\lambda t} x_0 + \sigma \int_0^t e^{\lambda(s-t)} dW_s$

Then X_t solves the SDE:



$$dX_t = -\lambda X_t \, dt + \sigma \, dW_t.$$
$$X_0 = x_0$$

Ex 5.3 ~ Exponential Martingales

$$X_t = x_0 \exp\left\{ \int_0^t \alpha_s \, dW_s - \frac{1}{2} \int_0^t \alpha_s^2 \, ds \right\}$$

$$dX_t = X_t \alpha_t \, dW_t$$

$$\mathbb{E}[X_t] = x_0$$

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s.$$

Ex 5.4

$$dX_s = \alpha(s) \, ds + \psi(s) \, dW_s$$

deterministic

$$\Rightarrow X_t \sim N\left(x_0 + \int_0^t \alpha(s) \, ds, \int_0^t \psi(s)^2 \, ds \right)$$

$$f(\gamma) := \mathbb{E}\left[e^{X_t \gamma} \right].$$